SK-Adapter++

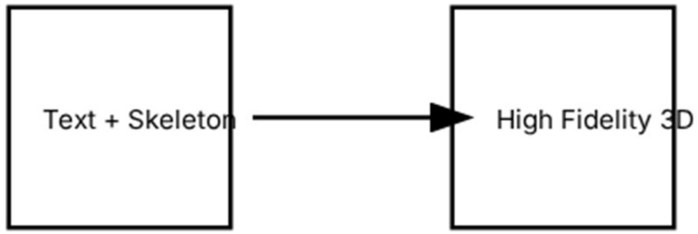
Enhancing Skeleton-Guided Native 3D Generation via Modality Compatible SK-Adapter

Ao, Chen, Liu, Wang

1. THE PROBLEM

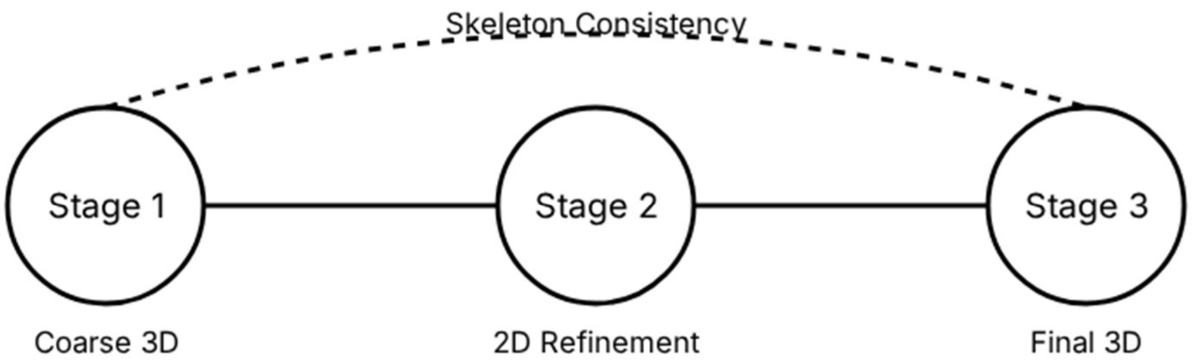
Quality vs. Controllability

- **Text-to-3D:** Structurally faithful (via skeletons) but limited visual quality (blurry, inconsistent).
- **Image-to-3D:** High visual fidelity but lacks structural control for specific poses.
- **The Gap:** Users need a way to combine precise 3D articulation with high-fidelity textures without pre-aligned image-skeleton pairs.



Goal: Bridging the Trade-off

2. OUR APPROACH



SK-Adapter++ Pipeline: Iterative refinement using a newly trained **image-conditioned SK-Adapter** that combines DINOv2 features with structural tokens.

3. BASELINES & METRICS

Metric Type	Metric Name	Measurement Goal
Structural	ReRigging Score (↓)	Chamfer distance between extracted and input skeleton.
Semantic	CLIP Score (↑)	Cross-modal similarity between renders and prompt.
Quality	PickScore (↑) / KD-DINO	Perceptual quality and distributional visual fidelity.

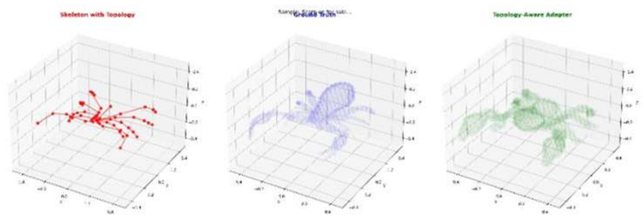
Baselines: SK-Adapter (Text only), SKDream (2D-projection), and Trellis (Image-to-3D).

4. ACHIEVEMENTS SO FAR

100%

Model Training Complete

- **Image-conditioned Adapter:** Successfully trained on Objaverse-TMS dataset.
- **Modality Compatibility:** Successfully injected skeleton tokens into image-conditioned Trellis backbone.



5. WHAT HAS NOT WORKED?

Topological Confusion

Models sometimes confuse anatomically similar classes. For example, a **Scorpion** might be generated with **Spider** leg topologies.

Semantic Orientation

The "Head" and "Tail" of articulated creatures are occasionally swapped during the 2D refinement stage (Stage 2).

Key Learning: LLM reasoning is required to verify semantic parts before the final image-conditioned generation stage.

6. PLANNED NEXT STEPS

While the **models** are trained, the **automated workflow** remains to be built.

Workflow Integration

Build the end-to-end Python pipeline connecting Stages 1, 2, and 3.

LLM Verifier

Implement an LLM-based check to fix semantic orientation errors.

Final Benchmark

Conduct a full comparative analysis against text-only SK-Adapter.

7. FEEDBACK & Q&A

Any Questions?